

Switching to direct feed instead of document camera next week

Anonymous feedback survey in 2 ± 1 weeks

Lecture quiz tomorrow, don't forget

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5: Line through $(3, 4, 0)$ perpendicular to $\langle 2, 2, 0 \rangle$ and $\langle 0, 1, 1 \rangle$.

(2) Recall: Point on line + direction vector of line is enough to get equation for line.

Let $\vec{v}$ be the direction, then $\vec{v} \perp$

$\langle 2, 2, 0 \rangle, \langle 0, 1, 1 \rangle$, so we can take

$$\vec{v} = \langle 2, 2, 0 \rangle \times \langle 0, 1, 1 \rangle = \langle 2, -2, 2 \rangle.$$

$$\begin{array}{l} 2 \ 2 \ 0 \\ 0 \ 1 \ 1 \end{array} \quad \begin{array}{l} i: 2 \cdot 1 - 0 \cdot 0 = 2 \\ j: -(2 \cdot 1 - 0 \cdot 0) = -2 \\ k: 2 \cdot 1 - 2 \cdot 0 = 2 \end{array}$$

Since we only care about direction and not magnitude,

$\langle 1, -1, 1 \rangle$ also works.

The line passes through $(3, 4, 0)$, so

$\langle 3, 4, 0 \rangle + t\langle 1, -1, 1 \rangle = \langle 3+t, 4-t, t \rangle$ is
vector equation of the line.

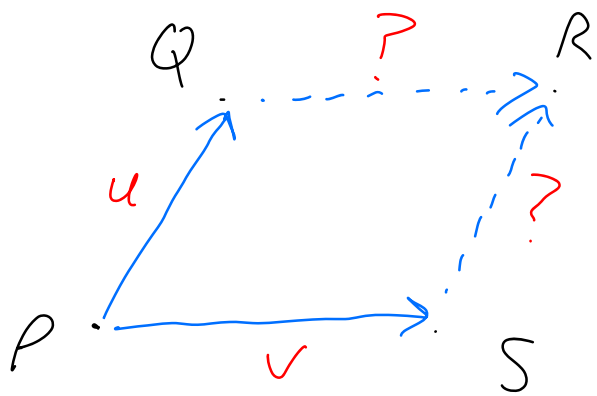
Convert to symmetric:

$$\frac{x-3}{1} = \frac{y-4}{-1} = \frac{z-0}{1}$$

(b) Recall yz -plane is plane $x=0$, so let's
set $x=0$ to find intersection. $x=3+t$,
so $t=-3 \Rightarrow y=4-t=7, z=t=-3$,
so intersection is $(0, 7, -3)$.

3: $P=(1, 2, 1), Q=(2, 5, 4), R=(6, 9, 12), S=(5, 6, 9)$

(2) Let's show that $PQRS$ is a parallelo-
gram, then this would mean P, Q, R, S are
coplanar, which means $\vec{PQ}, \vec{PR}, \vec{PS}$ coplanar.



Let $u = \overrightarrow{PQ}$, $v = \overrightarrow{PS}$.

2 opposite sides form
a parallelogram if and
only if they are

represented by the same vector, so we
need to show $\overrightarrow{SR} = u$ as well.

$$\vec{u} = Q - P = (2, 5, 4) - (1, 2, 1) = \langle 1, 3, 3 \rangle$$

$$\vec{v} = S - P = (5, 6, 9) - (1, 2, 1) = \langle 4, 4, 8 \rangle,$$

$$\text{and indeed } \overrightarrow{SR} = \langle 6, 9, 12 \rangle - \langle 5, 6, 9 \rangle = \vec{u}$$

(b) Recall a parallelogram spanned by two
vectors u & v has area $\|u \times v\|$.

$$u \times v = 4 (\langle 1, 3, 3 \rangle \times \langle 1, 1, 2 \rangle) = 4 \langle 3, 1, -2 \rangle,$$

$$\begin{array}{l} 133 \\ 112 \end{array} \quad \begin{array}{l} \therefore 3 \cdot 2 - 1 \cdot 3 = 3 \\ \therefore -(2 \cdot 3) = -6 \\ \therefore 1 \cdot 3 - 3 \cdot 1 = 0 \end{array} \quad \text{so area} = \|u \times v\|$$

$$= 4 |\langle 3, 1, -2 \rangle| = \boxed{4\sqrt{14}}$$

(c) PQS is $\frac{1}{2}$ of the parallelogram,

$$\text{so } [PQS] = \frac{1}{2} \cdot 4\sqrt{14} = \boxed{2\sqrt{14}}$$

2c: 2 unit vectors orthogonal to a & b .

$$a = \langle 2, 1, -3 \rangle, \quad b = \langle 0, -2, 1 \rangle, \quad \text{and}$$

$\begin{array}{rcl} 2 & 1 & -3 \\ 0 & -2 & 1 \end{array}$	$\begin{array}{l} i: -5 \\ j: -2 \\ k: -4 \end{array}$	$a \times b = \langle -5, -2, -4 \rangle$ from previous parts of the problem. $\ a \times b\ = \sqrt{25 + 20} = \sqrt{45} = 3\sqrt{5},$
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so $u = \frac{a \times b}{\|a \times b\|} = -\frac{1}{3\sqrt{5}} \langle 5, 2, 4 \rangle$ is

a unit vector orthogonal to a & b .

Also, $-u$ will work since u does,

so $\pm \frac{1}{3\sqrt{5}} \langle 5, 2, 4 \rangle$ are 2 such vectors.

6: Parallel, intersecting, or skew

(2) Recall lines are parallel iff direction vectors are parallel.

$L_1: \langle -1, 2, 3 \rangle + \langle 3, 4, -2 \rangle t$, so its direction is $\langle 3, 4, -2 \rangle$

L_2 : from denominators, direction is $\langle 2, -3, -3 \rangle$

These 2 vectors aren't parallel, so L_1 & L_2 aren't parallel.

To check if 2 lines intersect, you can plug equation of 1st into 2nd.

1st line has $x = -1 + 3t$, $y = 2 + 4t$, $z = -2t$,

2nd line has $\frac{x-1}{2} = \frac{y}{-3} = \frac{z+1}{-3}$.

$$\frac{-1+3t-1}{2} = \frac{2+4t}{-3} \Rightarrow$$

$$-3(-2+3t) = 2(2+4t) \Rightarrow 4+8t = 6-9t$$

$$\Rightarrow 17t = 2 \Rightarrow t = \frac{2}{17}$$

$$\frac{2+4t}{-3} = \frac{y}{-3} = \frac{z+1}{-3} = \frac{4-2t}{-3}$$

$$\Rightarrow 2+4t = 4-2t \Rightarrow 6t = 2$$

$$\Rightarrow t = \frac{1}{3}$$

$\frac{2}{17} \neq \frac{1}{3}$, so can't satisfy both symmetric equations simultaneously, so no sol. $\Rightarrow L_1$ & L_2 don't intersect $\Rightarrow$ skew.